Clinical Measurement Audits and Two-Phase Imputation: Concise Study Guide

A twelve-lesson study guide to Patrick Iben's *Statistical Inference and Two-Phase Audit Design for Contaminated Measurements in Clinical Research: The catmech R Package*, canonical manuscript paper/manuscript.Rmd (350 lines, SHA-256 e25e6b44561b4c5b8529d87867eda249665ab2269aa406e8217a90e3639c0758) and compiled 16-page PDF paper/manuscript.pdf (SHA-256 79c865d1e4f4ef216599aeb7b4ce931c237bb89b2c525c5b97d50ea6f8b56df1).

This educational companion explains the statistical definitions, benchmark archives, and R workflows in the frozen catmech manuscript; it does not establish universal interval coverage under unmodeled misspecification, validate per-record error probabilities for individual patient adjudication on diffuse real-world data, or certify FDA 21 CFR Part 11 or ICH E9 regulatory compliance.

How to use this twelve-lesson route

This concise guide is designed for clinicians, trial statisticians, and data auditors who want the core mathematical logic, empirical findings, R function calls, and scope boundaries in twelve short lessons (S1 through S12). You can also step through these exact twelve lessons interactively in an R console by running `source('education/interactive_tutor.R'); catmech_tutor()`, or read the full thirty-unit passage-by-passage companion in COMPANION.md (PDF).

S1. The three-way clinical audit dilemma and why recorded values matter

Where we are

Abstract and Section 1, manuscript lines 1-41 (PDF pages 1-2).

What the passage is doing

When an audit re-measures a subset of records (`R_i`), trialists face three flawed conventional choices: (1) naively trusting recorded values (`W_i`) causes attenuation bias; (2) analyzing only audited rows (`gold_only`) inflates standard errors; and (3) setting unaudited `W_i` to `NA` (`reference_normal`) wastes accurate recordings. On ClinicalTrials.gov (`N = 27,208`) and NHANES (`N = 3,908`), keeping `W_i` via `catmech_reference(use_recorded = TRUE)` reduces RMSE by 43% to 48% over discarding unaudited values.

In ordinary language

Audits check a small slice of records against a trusted reference. Keeping recorded numbers cuts error nearly in half compared with throwing unchecked numbers away.

Evidence status

SCOPE and MEASURED_DATED summary across 10 frozen benchmark archives.

Do not accidentally conclude

Do not conclude that standard MAR imputation (`mice`, `Amelia`) is biased when unaudited `W_i` values are discarded; its penalty in that mode is wider confidence intervals.

Try it

Run `round(100 * (1 - 0.0409 / 0.0785), 1)` in R to verify the NHANES RMSE reduction at a 25% audit. **Answer:** `47.9`%.

S2. Two-phase contamination mixture model and marginalized corruption probabilities

Where we are

Sections 2.1-2.2, manuscript lines 42-70 (PDF pages 2-3).

What the passage is doing

Latent true values follow `T_i | X_i ~ N(X_i beta, sigma_T^2)` and recorded values follow `W_i = (1 - delta_i) T_i + delta_i (T_i + epsilon_i)` with `delta_i in {0, 1}` and `epsilon_i ~ N(mu_e, sigma_e^2)`. On unaudited rows, `catmech_contamination()` integrates out `(delta_i, T_i)` to compute the exact marginal corruption probability `p_i`. When `delta_i = 0`, `T_i = W_i` deterministically; when `delta_i = 1`, `T_i` is drawn from the precision-weighted conjugate Gaussian posterior.

In ordinary language

Each row has a hidden true value and a recorded value. The mixture model compares how well an unchecked recording fits the clean curve versus the wider error curve.

Evidence status

DEFINITION and ANALYTIC conditional posteriors for `catmech_contamination()`.

Do not accidentally conclude

Do not conclude that `reference_sd` (`sigma_ref`) is estimated from the data; it defaults to `0` (exact recounts) and is a user-declared sensitivity constant.

Try it

When `delta_i = 0` is drawn for an unaudited record in `catmech_contamination()`, what is the imputed value `T_i`? **Answer:** `T_i = W_i` deterministically.

S3. Gaussian reference imputation in C++17: use_recorded = TRUE versus FALSE

Where we are

Section 2.2, manuscript lines 71-72 (PDF page 3).

What the passage is doing

`catmech_reference()` implements fast conjugate Gaussian reference imputation in C++17 (`RcppArmadillo`). With `use_recorded = TRUE` (Simple Linear MI, `cole_mi`), `W_i` enters directly as a continuous linear calibration predictor alongside `X_i` (`T_i | W_i, X_i`). With `use_recorded = FALSE` (`reference_normal`), unaudited `W_i` values are discarded.

In ordinary language

When errors are small and spread out, a fast linear regression on the audited rows calibrates the unchecked numbers without needing a mixture model.

Evidence status

DEFINITION of `catmech_reference(use_recorded = TRUE)` versus `use_recorded = FALSE`.

Do not accidentally conclude

Do not forget to pass `use_recorded = TRUE` explicitly when you want Simple Linear MI; the formal function default is `use_recorded = FALSE` for v0.3 backward compatibility.

Try it

In `Lab C16` (`education/run_companion_labs.R`), why does `use_recorded = TRUE` yield a smaller pooled standard error than `use_recorded = FALSE`? **Answer:** Because `W_i` explains substantial residual variance in `T_i` on unaudited rows.

S4. Rough two-phase audit-size planning and the NO-GO planner boundary

Where we are

Section 2.3, manuscript lines 73-84 (PDF pages 3-4).

What the passage is doing

At audit fraction `f = n_audit / N`, the confidence interval width ratio relative to a 100% census audit is `rho(f) = sqrt(1 + k (1 - f) / f)`. `catmech_audit_power()` is a rough, unvalidated planning aid: across `2,000` log-normal draws of `k_hat` it reports the share of plausible `k` values that meet `rho <= rho_target`. The output calls that share `power`, but it is not the power of a hypothesis test. Without a pilot plan its default `k` is optimistic, so it also reports the audit size that linear imputation needs at the same correlation, often several times larger. In pre-specified validation (`validation/audit-plan-study`), `catmech_audit_plan()` received a `NO-GO` verdict (`11.3%-16.8%` median width error at `f = 5%`) and is documented as exploratory.

In ordinary language

The audit-size tool makes a rough guess from assumed error rates, and nobody has checked its answers. The empirical pilot planner missed its 10% accuracy target and stays exploratory.

Evidence status

ANALYTIC influence-function ratio and MEASURED_DATED `NO-GO` boundary on `catmech_audit_plan()`.

Do not accidentally conclude

Do not read the `power` column of `catmech_audit_power()` as the power of a test, and do not treat `catmech_audit_plan()` as a validated regulatory sample-size guarantee; its pre-specified status is `NO-GO`.

Try it

Compute `rho(0.25) = sqrt(1 + 0.27 * (1 - 0.25) / 0.25)` in R (`round(sqrt(1 + 0.27 * 3), 4)`). **Answer:** `1.3454`.

S5. Rubin combining rules and Barnard-Rubin finite-sample degrees of freedom

Where we are

Section 2.4, manuscript lines 85-94 (PDF page 4).

What the passage is doing

Across `m` completions, total variance is `T = U_bar + (1 + 1/m) B` and the fraction of missing information is `lambda_hat = (1 + 1/m) diag(B) / diag(T)`. To prevent anti-conservative intervals when sample size `N`, site count `C`, or event count `d` is small, `catmech` computes Barnard-Rubin (1999) degrees of freedom `nu_BR = (nu_m * nu_obs) / (nu_m + nu_obs) < nu_com`.

In ordinary language

Old imputation formulas can claim more degrees of freedom than the study has sites or events. The Barnard-Rubin formula caps degrees of freedom below the complete-data limit.

Evidence status

ANALYTIC Rubin (1987) and Barnard-Rubin (1999) pooling formulas.

Do not accidentally conclude

Do not assume that increasing `m` to 100 lets `nu_BR` exceed `C - 1` in a clustered study; `nu_BR` is strictly bounded below `nu_obs < nu_com = C - 1`.

Try it

For `C = 10` clinical sites (`nu_com = 9`), `m = 5`, and `lambda_hat = 0.2308`, what is `nu_BR` (`Lab C10-C11`)? **Answer:** `5.3577` (strictly below `9`, whereas classical `nu_m = 75.11`).

S6. Multi-endpoint pooling across linear (CV3), GLM (CR1S), and Cox PH (Lin-Wei) models

Where we are

Section 2.4, manuscript lines 95-100 (PDF pages 4-5).

What the passage is doing

`catmech_pool_lm()` uses the delete-one-cluster jackknife (`cv3`) with `nu_com = C - 1` when clustered; `catmech_pool_glm()` uses the CR1S Huber-White-Liang-Zeger sandwich with `nu_com = C - 1`; and `catmech_pool_coxph()` uses event-based `nu_com = d - p` when unclustered or the Lin-Wei score-residual sandwich with `nu_com = C - 1` when clustered.

In ordinary language

Linear models use a leave-one-site-out jackknife variance, logistic and count models use a small-sample sandwich, and survival models count observed events.

Evidence status

ANALYTIC multi-endpoint covariance and degrees-of-freedom specifications.

Do not accidentally conclude

Do not use total sample size `N - p` for unclustered Cox proportional hazards degrees of freedom; `catmech_pool_coxph()` uses observed failure events `d - p`.

Try it

In an unclustered Cox PH model with `N = 600` patients, `d = 45` events, and `p = 3` predictors, what is `nu_com`? **Answer:** `d - p = 45 - 3 = 42`.

S7. Out-of-sample rank-PIT calibration checking and its slope-bias blind spot

Where we are

Section 2.5, manuscript lines 101-106 (PDF page 5).

What the passage is doing

`catmech_holdout_check()` runs K-fold cross-validation on audited rows and computes tie-randomized rank PIT values `u_i` against `Uniform(0, 1)`. While false alarms were `0/40` under the true model, alarm power against structural misspecification was low (`0/40` in `nonlinear_target`, `1/40` in `heavy_tail_target`) because held-out rows test observation-level prediction rather than regression slope bias.

In ordinary language

Holdout checking hides part of the audit and tests if row predictions match the hidden truth. Passing this screen does not guarantee an unbiased regression slope.

Evidence status

ANALYTIC rank-PIT formula and MEASURED_DATED diagnostic power boundary.

Do not accidentally conclude

Do not treat `miscalibration_detected = FALSE` in `catmech_holdout_check()` as proof that downstream regression slopes are unbiased.

Try it

Why does `catmech_holdout_check()` randomize exact ties (`T_{i,-k}^{(j)} = R_i^*`) using `U_i ~ Uniform(0, 1)`? **Answer:** Because the contamination mixture places a discrete point mass at `W_i` on uncorrupted rows; without tie randomization, exact matches would spike and trigger false KS alarms.

S8. Synthetic contamination benchmarks (52,950 requests) and three stress-cell failures

Where we are

Sections 3-5.1, manuscript lines 107-249 (PDF pages 5-10).

What the passage is doing

Across 52,950 synthetic requests, catmech_contamination() narrowed CIs by 44.3% over complete-case analysis in missed_flags (0.1842 vs 0.3309) and prevented the -0.2354 linear-MI bias under gross 10x outliers (error_gross_and_small, 94.0% vs 64.7% coverage). However, it failed in three stress cells: heavy_tail_target (87.3% coverage), nonignorable_missed (79.3% coverage), and noisy_undeclared (88.7% coverage, restored to 91.8% when reference_sd is declared).

In ordinary language

Under synthetic gross or unflagged errors, the mixture model stays unbiased and cuts interval width by 44%. It fails under heavy tails, non-ignorable selection, or undeclared reference noise.

Evidence status

MODELED_LOCKED synthetic benchmark results across 27 scenarios.

Do not accidentally conclude

Do not use catmech_contamination() when the true distribution T_i has heavy t_4 tails (heavy_tail_target); use catmech_reference(use_recorded = TRUE) (92.0% coverage) instead.

Try it

What caused catmech_contamination() to under-cover (87.3%) in heavy_tail_target?
Answer: The Gaussian mixture misclassified genuine heavy-tailed extremes in T_i as recording errors and shrunk them toward X_i beta.

S9. Clustered variance (148,500 requests): CV3 jackknife and predictor-slope attenuation

Where we are

Section 5.2, manuscript lines 250-258 (PDF page 10).

What the passage is doing

Across 148,500 clustered requests on simulated cohorts and 11 Rwanda HMIS indicators, cv3 beat cr1s and cr2 (which inflated widths up to 95%). However, when the audited variable T_i was a **predictor** (Y ~ T + x + stratum), clustered random-intercept imputation (catmech_reference(cluster =)) attenuated slopes by up to -16.1% (84.4% coverage), whereas fitting catmech_reference(use_recorded = TRUE) without cluster and pooling with catmech_pool_lm(cluster =) achieved -3.0% bias and 95.4% coverage.

In ordinary language

Across clinical sites, the leave-one-site-out jackknife gives the best interval coverage. However, random-intercept imputation shrinks slopes when the audited variable is a predictor.

Evidence status

MEASURED_DATED and MODELED_LOCKED clustered variance and multilevel MI evaluation.

Do not accidentally conclude

Do not pass `cluster` to `catmech_reference()` when the audited variable is a predictor in the downstream regression (`catmech_pool_lm()` warns if you do).

Try it

What is the recommended two-step R call when an audited biomarker is a predictor in a multicenter linear regression? **Answer:** Fit `catmech_reference(..., use_recorded = TRUE)` without `cluster`, then pool with `catmech_pool_lm(..., cluster = 'site')`.

S10. Public paired-discrepancy benchmarks (33,800 requests) and near-chance row error AUCs

Where we are

Section 6.1, manuscript lines 259-279 (PDF page 11).

What the passage is doing

On ClinicalTrials.gov (`N = 27,208`) and NHANES 2017-2018 (`N = 3,908`), retaining recorded values reduced RMSE by `43%` to `48%` over discarding unaudited values. However, because real-world discrepancies were small relative to `T_i | X_i`, per-record corruption probabilities `p_i` had near-chance discrimination (`AUC = 0.543` on ClinicalTrials.gov and `0.500` on NHANES, versus `0.962` in synthetic `missed_flags`).

In ordinary language

On real data with diffuse errors, keeping recorded numbers cuts slope error nearly in half, but per-row error probabilities score no better than a coin flip.

Evidence status

MEASURED_DATED empirical benchmark across ClinicalTrials.gov and NHANES.

Do not accidentally conclude

Never use individual record-level corruption probabilities `p_i` from `catmech_contamination()` to adjudicate or discard specific patient records in diffuse real-world data.

Try it

What was the ROC-AUC of per-record corruption probabilities `p_i` at a 25% audit on ClinicalTrials.gov and NHANES? **Answer:** `0.543` on ClinicalTrials.gov and `0.500` on NHANES (near coin-flip chance `0.500`).

S11. Real-audit slope recovery (45,600 requests) and the methodological decision tree

Where we are

Sections 6.2-8, manuscript lines 280-328 (PDF pages 12-14).

What the passage is doing

On Rwanda HMIS facility recounts (`45,600` requests), Rule 2 comparing `catmech_contamination()` against `Linear MI` was `INCONCLUSIVE`: the mixture had lower

point RMSE on PNC (0.053 vs 0.072) and Iron (0.112 vs 0.156), but failed 95% coverage on all three primary slopes (71.6% on PNC, 66.6% on Syphilis, 35.4% on Iron) and failed the >= 90% return rate on two. Therefore, Section 7 recommends catmech_contamination() only for gross, separable outliers and catmech_reference(use_recorded = TRUE) for moderate or diffuse real-world discrepancies.

In ordinary language

On Rwanda health-center recounts, the mixture model had lower point error on two slopes but failed interval coverage. For everyday moderate errors, use simple linear imputation.

Evidence status

MEASURED_DATED real-audit evaluation (INCONCLUSIVE Rule 2) and INTERPRETIVE decision tree.

Do not accidentally conclude

Do not claim that catmech_contamination() outperformed Simple Linear MI on real-world audit intervals; its interval advantage remains synthetic-only.

Try it

Under the Section 7 decision tree, what two checks must pass before trusting catmech_contamination() on a dataset with suspected gross errors? **Answer:** A >= 90% interval return rate and positive held-out Brier/RMSE skill over raw recorded values in catmech_holdout_check().

S12. Provenance reporting (catmech_audit_trail), software QA, and regulatory boundaries

Where we are

Sections 9-12, manuscript lines 329-350 (PDF pages 14-16).

What the passage is doing

catmech_audit_trail() outputs a provenance record (Markdown, JSON, or HTML) recording R/OS metadata; SHA-256 hashes of the input data and all m completed datasets, computed over the serialized R data frames, and of the assumption specification as canonical JSON text; MCMC convergence status; and pooled tables. A rerun can be compared with the hashes, but they do not cover the pooled results or make the record tamper-evident. This supports documentation under FDA 21 CFR Part 11 and ICH E9 without certifying compliance. The package is MIT-licensed and verified by 199 testthat tests with 1,812 expectations (0 failures/warnings/skips), R CMD check --as-cran (0 errors/warnings), and 500,000+ replicate checks.

In ordinary language

The audit trail function writes a report with SHA-256 data hashes, model settings, and convergence status. A rerun can be compared with it, but it does not prove the data were untouched or certify legal compliance.

Evidence status

DEFINITION, SCOPE, and MEASURED_DATED software verification summary.

Do not accidentally conclude

Do not represent `catmech_audit_trail()` as a legal compliance certification or a tamper-evident record; regulatory compliance depends on the surrounding validated system, access controls, and procedures, and anyone who edits the data can regenerate the record.

Try it

Run `Lab C28` in `education/run_companion_labs.R`. What default title and footer heading does `catmech_audit_trail()` print? **Answer:** Default title `Catmech Audit Trail & Provenance Report` and footer heading `### Audit Trail & Provenance Record`.